

Maximus Jialong Niu

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TECHNICAL SKILLS

Languages: JavaScript/TypeScript, Python, SQL, C, C++

Frameworks & Libraries: React, Next.js, Tailwind, Storybook, Node.js, Flutter

Databases: PostgreSQL (pgvector), DynamoDB

Technologies: Linux, Git, AWS (S3, IAM, Lambda), Azure (DevOps, Web Apps, AD B2C), Docker, Jest, Supabase

EMPLOYMENT

React Developer

Mar. 2025 – Present

Venuiti Healthcare

Waterloo ON, Canada

- Led the frontend development of the myMedcan healthcare portal using **Next.js** and **TypeScript**, integrating a **Salesforce**-based backend that supports **20,000+** registered users
- Engineered a core multi-step clinical appointment booking and purchasing flow, utilizing **XState** and **React Query** for state management, driving **15,000** bookings and **\$168,000** in purchases monthly
- Integrated **Azure AD B2C JWT** authentication with custom user flows and middleware to eliminate client-side PHI exposure, ensuring **100% HIPAA** and **PIPEDA** compliance and route-level access control
- Built a library of **199+** reusable **React** components and hooks with **Tailwind** and **Storybook**, achieving **WCAG 2.2 AA** accessibility compliance across all components through automated accessibility testing
- Architected an **Azure DevOps CI/CD** pipeline that containerized the myMedcan frontend with **Docker**, ran a custom **Jest** unit test suite with **80%** code coverage, and enabled a complete build-test-deployment cycle across development, staging, and production environments on **Azure Web Apps** in under **5 minutes**

Software Developer

Jan. 2023 – Apr. 2023

Wind River Systems

Ottawa ON, Canada

- Wrote and executed **50+** **C/C++** unit tests on a simulated **x86** board to validate Unix utility functions ported to the **VxWorks RTOS**, ensuring **POSIX**-compliant behavior across different embedded environments
- Built an internal **Python** and **Flask** automation tool to convert between Markdown formats, processing **500+** pages and streamlining documentation workflows across teams

Full Stack Developer

Sep. 2020 – Dec. 2020

Syntu Software

Waterloo ON, Canada

- Architected a serverless system from scratch for handling sensitive legal evidence using **AWS DynamoDB**, **S3**, **Lambda**, and **IAM**, ensuring **100% PIPEDA** compliance

Software Developer

Jan. 2020 – May. 2020

The Co-operators

Guelph ON, Canada

- Built an automation workflow utilizing **OCR** and **web scraping** to test an internal policy filing application, achieving **400+** policy submissions per hour

PROJECTS

ReWatched – Semantic search over your entire YouTube watch history

- Engineered an **MV3 Chrome extension** to enable content-based semantic search across YouTube watch history; performs background indexing by fetching video transcripts, breaking them into overlapping chunks for embedding with **OpenAI**, and persisting vectors locally in **IndexedDB**
- Implemented a fully client-side query pipeline using **cosine similarity** over stored embeddings, returning ranked results with match confidence, thumbnails, and timestamp deep links, achieving **0.95 recall@10** across **50** test queries on a **500-video** dataset

RoomEase – AI-powered co-living assistant mobile app

- Built a cross-platform **Flutter** app with an **agentic AI** assistant that routes user intent to structured actions (chore delegation, message relay, schedule management) via a multi-stage preprocessing pipeline
- Managed conversation context using **OpenAI** and **Pinecone**, retrieving the top 10 relevant messages per query and achieving a **3.4s** agentic response time and **55ms** DB reads in benchmarks

EDUCATION

University of Waterloo

Apr. 2024

Bachelor of Applied Science, Honours Computer Engineering

Relevant Courses: Software Architecture, Database Systems, Machine Learning, Systems Programming and Concurrency, Computer Networks, Computer Security, Compilers, Algorithms & Data Structures